PAX SERVICES GROUP

PAXSEAL™ BUILDING LIFECYCLE PLAN

Roof Lifecycle Cost Analysis

MARKET PROJECTION
Typical school

Hampton Roads market
October 14, 2026

Planned maintenance is projected to save the owner of a typical Hampton Roads school $1.34M over the next 20 years.

About 35% less than reactive repair: $990K lower spending plus $352K more roof life at year 20. Range $693K–$1.75M from conservative to upside assumptions; $698K in today's dollars at a 7% discount rate.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE · CONDITION	NEW ROOF TODAY
Hampton Roads	K-12	120,000 sq ft	Modified bitumen	8 yrs · Good	$960K

PAXSEAL NET COST
$2.48M
$2.39M spent + $576K roof used − $480K roof left

REACTIVE NET COST
$3.83M
$3.38M spent + $576K roof used − $128K roof left

REPAIR & STORM SAVINGS
$1.63M
Unbudgeted repairs and storm damage avoided

ROOF PRICES
2× in 15 yrs
At 4.6%/yr, the long-run trend in commercial roofing costs

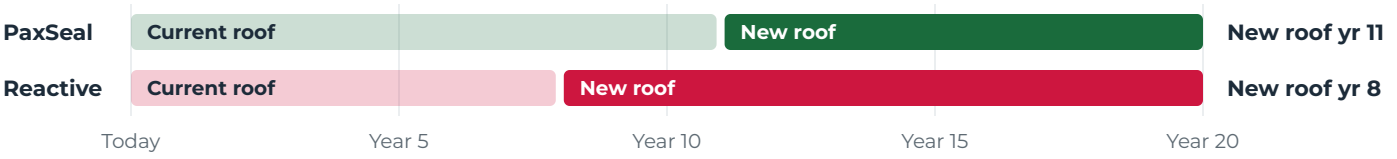
Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal fees	$441K	—	**−$441K**
Repairs paid by owner	$234K	$1.79M	**+$1.56M**
Expected storm damage	$140K	$210K	**+$70K**
Roof replacements	$1.57M	$1.38M	**−$190K**
Cash spending	**$2.39M**	**$3.38M**	**+$990K**
Current roof value used	$576K	$576K	—
Roof life left at year 20	−$480K	−$128K	**+$352K**
Net 20-year cost	**$2.48M**	**$3.83M**	**+$1.35M**

What this means

1. **A later replacement.** The next new roof moves from year 8 to year 11, leaving 10 years of roof life at year 20 instead of 2.
2. **Smaller repairs.** Emergency repairs run about 3.9× the planned rate in a school, and the PaxSeal fee covers half of planned repairs. Together that saves $1.56M.
3. **A predictable budget line.** Emergency repairs become a fixed annual cost the board can plan for, and leaks stop closing classrooms.
4. **The program pays for itself.** Each $1 of fees avoids about $3.25 of other cash roof costs.

Local risk profile

Norfolk city, VA (51710)

Hurricane / tropical-storm wind

About 4.8 tropical storms or stronger within 50 nautical miles of Norfolk per 20 years (NOAA HURDAT2 1975–2024). Expected storm damage to this roof over 20 years is **$140K** if maintained and **$210K** with reactive repair only, before insurance. A documented maintenance history also helps when an insurer reviews a claim.

Hail	**Relatively Low**
Strong wind	**Relatively Moderate**
Tornado	**Relatively Moderate**
Hurricane	**Relatively Moderate**
Winter weather	**Relatively Moderate**
Ice storm	**Relatively Moderate**

FEMA National Risk Index rating

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	**20 / 15 yrs**	Maintained / reactive. Industry life tables; maintenance adds ~5 yrs (NRC Canada, warranties)
Roof cost growth	**4.6%/yr**	BLS Producer Price Index, nonresidential roofing, 2007–2025
Replacement cost	**$8.00/sq ft**	Hampton Roads contractor price ranges
Reactive repair cost	**3× planned × 1.3**	Industry estimates; ×1.3 for leak damage (school). PaxSeal fee covers half of planned repairs
Storm damage	**0.44%/yr**	Of roof cost, before insurance; ×1.5 if reactive. FEMA National Risk Index
PaxSeal fee	**$16,400/yr**	Year 1, rising 3%/yr. Estimate; your proposal states the actual fee

Next step: a roof assessment and a written PaxSeal proposal for this building. Call PAX Services Group at 1-855-729-7510 or email info@paxserv.com.

Estimates, not a quote. Projections from Hampton Roads market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20; both roof-value figures are in today's dollars. Storm damage is before insurance.